HOMEWORK ASSIGNMENT 7 – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

QUESTION 1: Picture It!

In Quiz 05, we briefly explored a 2025[1] study, designed to explore the data visualization course offerings across several highly-ranked (as determined by US News) US higher-education institutes. In this problem, we'll explore the data a bit further. In particular, 50 of the 154 institutes included in the sample are classified as "liberal arts colleges" whereas the remaining 104 are classified as "universities." Among the liberal arts colleges, 16 offered at least one data visualization course and the remaining 34 did not. Among the universities, 78 offered at least one data visualization course and the remaining 26 did not.

(a) Fill in the following two-way table with the appropriate values.

	Liberal Arts College	University	Total
At Least One Data Viz. Course Offered	16	78	94
No Data Viz. Course Offered	34	26	60
Total	50	104	154

(b) Calculate the conditional proportion of liberal arts colleges that offer at least one data visualization course, and interpret this value.

Solution: $\frac{16}{50} = \mathbf{0.32 = 32\%}$. That is: 32% of the 50 sampled liberal arts colleges offered at least one data visualization course.

(c) Sketch a segmented bar graph of the data contained in the two-way table.

Solution:

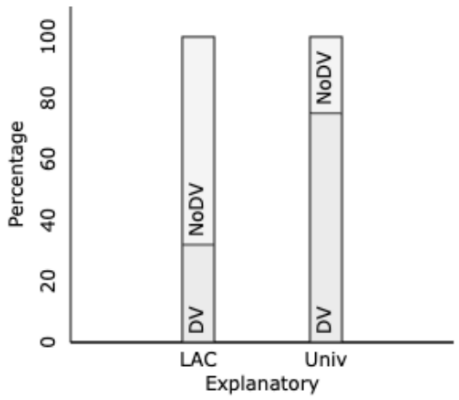


Figure 1: Segmented Bar Graph of the Data, Generated by the Two Proportions Applet

[1] Branson, Z., Paz Parra, M., & Yurko, R. (2025). The Landscape of College-Level Data Visualization Courses, and the Benefits of Incorporating Statistical Thinking. *Journal of Statistics and Data Science Education*, 33(4), 390–406. https://doi.org/10.1080/26939169.2025.2537049

(d) Sketch a mosaic plot of the data contained in the two-way table.

Solution:

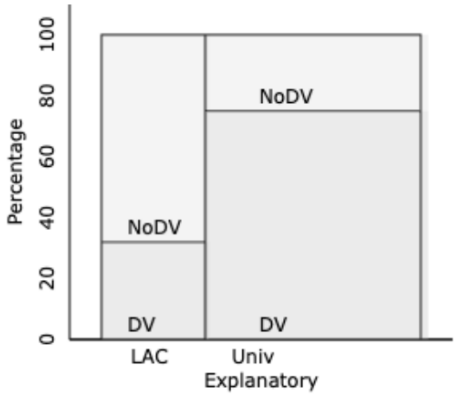


Figure 2: Mosaic Plot of the Data, Generated by the Two Proportions Applet

(e) Calculate the difference in conditional proportions of data visualization course offerings between liberal arts colleges and universities.

Solution: $(\hat{p}_{\text{LAC}} - \hat{p}_{\text{Univ}}) = \frac{16}{50} - \frac{78}{104} = 0.32 - 0.75 =$ **-0.43**. Note: you may have done the subtraction in the other order - that's fine too! Either way we cut it, we get the following conclusions (which are equivalent):

- The sample proportion of liberal arts colleges that offer at least one data visualization course is at 43 percentage points lower than the sample proportion of universities colleges that offer at least one data visualization course
  - This is using the order (LAC — Univ)
- The sample proportion of universities that offer at least one data visualization course is at 43 percentage points higher than the sample proportion of liberal arts colleges that offer at least one data visualization course
  - This is using the order (Univ — LAC)

(f) Calculate the relative risk, ensuring that your final answer is greater than 1, and interpret this value.

Solution: $\frac{\hat{p}_{\text{Univ}}}{\hat{p}_{\text{LAC}}} = \frac{0.75}{0.32} \approx$ **2.344**. The sample proportion of universities that offer at least one data visualization course is around 2.344 times larger than the sample proportion of liberal arts colleges that offer at least one data visualization course.

(g) Calculate and interpret the odds ratio comparing the odds of offering at least one data visualization course for the liberal arts colleges to the odds of offering at least one data visualization course for the universities. Interpret this value.

Solution:

- Odds of offering at least one data visualization course among Liberal Arts Colleges: 16/34

- Odds of offering at least one data visualization course among Universities: 78/26
- Odds ratio comparing the odds of offering at least one data visualization course among LA colleges to the odds of offering at least one data visualization course among Universities:

$$\text{odds ratio} = \frac{\left(\frac{16}{34}\right)}{\left(\frac{78}{26}\right)} \approx 0.1569 = \frac{1}{6.375}$$

The odds of offering at least one data visualization course among universities is around 6.375 times the odds of offering at least one data visualization course among liberal arts colleges.

QUESTION 2: Picture It! (Continued)

Consider again the setup of Question 1 above, along with the data described in it. In this question, we'll explore the following research question: are highly-ranked US universities more likely to offer at least one data visualization course than highly-ranked US liberal arts colleges? **Note:** below, we'll use "Univ." as an abbreviation for "University," and "LAC" as an abbreviation for "Liberal Arts College." Additionally, wherever necessary, use the order (University − LAC) and treat "success" as "offering at least one data visualization course."

(a) Calculate the value of the statistic (use the *difference* of sample proportions, and use University − Liberal Arts College).

Solution:

- Sample proportion of universities that offer at least one data visualization course: $\hat{p}_{\text{Univ}} = 78 \,/\, 104 = 0.75$
- Sample proportion of liberal arts colleges that offer at least one data visualization course: $\hat{p}_{\text{LAC}} = 16 \,/\, 50 = 0.32$
- Statistic: $(0.75 - 0.32) = \mathbf{0.43}$

(b) What are the appropriate symbols to represent the statistic?

- ◯ $\pi_{\text{Univ}} - \pi_{\text{LAC}}$
- √ $\hat{p}_{\text{Univ}} - \hat{p}_{\text{LAC}}$
- ◯ $\hat{p}_{\text{Univ}-\text{LAC}}$
- ◯ $\pi_{\text{Univ}-\text{LAC}}$

(c) Write down the null and alternative hypotheses using symbols and math only, but also provide a verbal description of any and all symbols you use.

Solution: Define:

- $\pi_{\text{Univ.}}$ to be the population proportion of all highly-ranked US universities that offer at least one data visualization course
- π_{LAC} to be the population proportion of all highly-ranked US liberal arts colleges that offer at least one data visualization course

The null hypothesis is that these two population proportions are equal, and the alternative is that $\pi_{\text{Univ.}} > \pi_{\text{LAC}}$ (note the way the research question is phrased). Therefore, we can write our null and alternative hypotheses as

$$\begin{cases} H_0: & \pi_{\text{Univ}} - \pi_{\text{LAC}} = 0 \\ H_A: & \pi_{\text{Univ}} - \pi_{\text{LAC}} > 0 \end{cases}$$

(d) Are the validity conditions met to conduct a theory-based test to answer the research question? Do not just state the validity conditions. Use the data from the sample to explain how the validity conditions are or are not met.

Solution: Becuase we are comparing two categorical variables, we need to ensure that all values in the cells of the two-way table are at least 10. Indeed:

- 16 liberal arts colleges offered at least one data visualization course
- 34 liberal arts colleges offered no data visualization courses
- 78 universities offered at least one data visualization course
- 26 universities offered no data visualization courses

and since all of these counts are greater than or equal to 10, the validity conditions **are** met.

(e) Use the Theory-based Inference applet to carry out a two-sample z-test to answer the research question.

Solution:

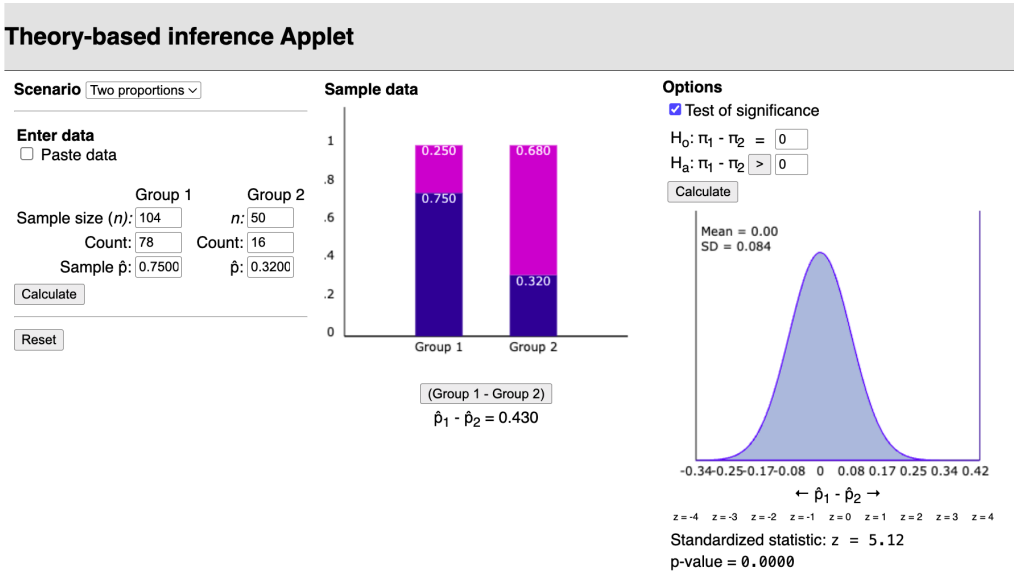


Figure 3: A Screenshot from the Theory-Based Inference Applet

The approximate theory-based *p*-value is 0.000.

(f) Based on the *p*-value you found, do you reject or fail to reject the null hypothesis?

- √ **Reject the null hypothesis in favor of the alternative hypothesis**
- ◯ Fail to reject the null hypothesis

(g) Given the results of your analysis, provide an overall conclusion **in context**. Be very clear what you can claim about the population proportions. IF you have evidence that one population proportion is greater/smaller/different than the other, then you need to state which one is greater/smaller/different.

Solution: Based on the statistic of 0.43 (the difference in sample proportion of institutes that offer at least one data visualization course, between the universities and the liberal arts colleges) we obtained a small p-value (0.000). This constitutes very strong evidence against the null hypothesis leading us to reject it in favor of the alternative. We do have very strong evidence that the population proportion of all highly-ranked US universities that offer at least one data visualization course is larger than the population proportion of all highly-ranked US liberal arts colleges that offer at least one data visualization course.

(h) To which population can you generalize the results of your study?

- ◯ We could generalize the results to all higher education institutes in the US
- ◯ We could generalize the results to higher-education institutes in the US that are currently (in 2026) highly ranked by US News
- √ **We could generalize the results to higher-education institutes in the US that were highly ranked by US News in 2025**

(i) Assuming we found a statistically significant result (which isn't necessarily the right answer to any of the previous parts!), could we make a **causal** claim? For instance, could we say that the classification of a higher-education institute (as a University or a Liberal Arts College) was the *cause* for offering or not offering a data visualization course, in 2025? Explain.

Solution: The key observation here is that this is an **observational study**, not a randomized experiment. As such, even though we found a statistically discernible difference in the population proportions described above, we **cannot make a causal claim**.

(j) Without using any applets, calculate a 90% confidence interval for the difference in the population proportions of Universities and Liberal Arts Colleges that offer at least one data visualization course.

Solution:

$$
\begin{aligned}
(\hat{p}_{\text{Univ}} - \hat{p}_{\text{LAC}}) &\pm 1.645 \cdot \sqrt{\frac{\hat{p}_{\text{Univ}}(1-\hat{p}_{\text{Univ}})}{n_{\text{Univ}}} + \frac{\hat{p}_{\text{LAC}}(1-\hat{p}_{\text{LAC}})}{n_{\text{LAC}}}} \\
&= (0.43) \pm 1.645 \cdot \sqrt{\frac{0.75(1-0.75)}{104} + \frac{0.32(1-0.32)}{50}} \\
&= (0.43) \pm 1.645 \cdot \sqrt{0.0018029 + 0.004352} \\
&= (0.43) \pm 1.645 \cdot \sqrt{0.0061549} \\
&= (0.43) \pm (0.0785) \\
&= \mathbf{(0.30, 0.56)}
\end{aligned}
$$

Note: In hindsight, I probably should have given you the multiplier here. Though, with that said, I think knowing that 1.645 is the multiplier for a 90% confidence interval is useful! But, as mentioned in lecture, you should DEFINITELY know the multiplier for a 95% confidence interval

(k) Use the Theory-Based Inference applet to calculate a 90% confidence interval for the difference in the population proportions of Universities and Liberal Arts Colleges that offer at least one data visualization course.

Solution:

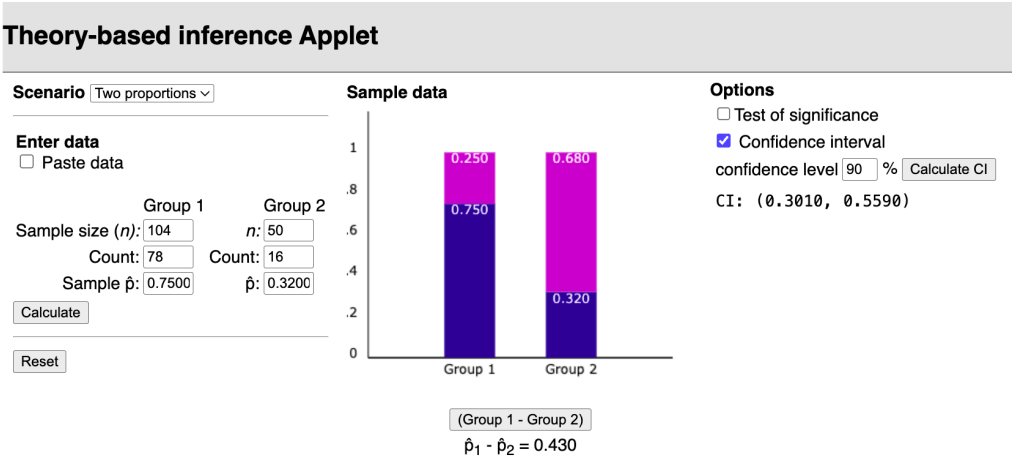


Figure 4: A Screenshot from the Theory-Based Inference Applet

The approximate 90% theory-based confidence interval, from the applet, is around **(0.3010, 0.5590)** which is very close to what we obtained using our formula.

(l) Interpret the interval you constructed in Part (k) above, using the **context** of the problem. Be sure to use words such as **greater** and/or **smaller** in your interpretation to make it clear if there is a population proportion higher than the other.

Solution: We are 90% confident that the population proportion of universities that offer at least one data visualization course is between 0.30 and 0.56 larger than the population proportion of all liberal arts colleges that offer at least one data visualization course, at least for higher-education institutes similar to the ones included in the study (highly ranked by US News, in the US, etc.)

QUESTION 3: What Are the Odds?

Imagine we conducted a randomized experiment in which subjects were administered one of two doses ("A" or "B") of a medicine, and observed to see whether or not the subject developed an adverse reaction ("yes" or "no"). The following two-way table was created from the data.

	Dose A	Dose B	Total
Adverse Reaction	20	40	**60**
No Adverse Reaction	30	50	**80**
Total	**50**	**90**	**140**

(a) Calculate the conditional proportion of subjects on Dose A that developed an adverse reaction.

Solution: $\frac{20}{50} = \mathbf{0.4 = 40\%}$

(b) Compute the relative risk of developing adverse reactions between the two doses, and interpret this value. **Hint:** remember that it's recommended to set up the relative risk to be greater than or equal to 1.

Solution:

- Conditional proportion of subjects on Dose A that developed an adverse reaction: 20/50 = 0.4
- Conditional proportion of subjects on Dose B that developed an adverse reaction: 40/90 = $0.\overline{4}$

Since the second quantity is larger than the first, we'll put it in the numerator:

$$\text{relative risk} = \frac{\text{risk of adverse effects under Dose B}}{\text{risk of adverse effects under Dose A}} = \frac{4/9}{4/10} = \frac{10}{9} = \mathbf{1.\overline{1}}$$

The risk of developing adverse effects under Dose B are about 1.11 times the risk of developing adverse effects under Dose A.

(c) Calculate the odds of not developing adverse reactions while on Dose A, and interpret this value.

Solution: $\frac{30}{20} = \mathbf{1.5}$. Under Dose A: for every patient that develops adverse reactions, there are 1.5 patients that don't develop adverse reactions.

(d) Calculate and interpret the odds ratio comparing the odds of developing adverse reactions under Dose A to the odds of developing adverse reactions under Dose B. Interpret this value.

Solution:

- Odds of developing adverse reactions under Dose A: 20/30
- Odds of developing adverse reactions under Dose B: 40/50
- Odds ratio comparing the odds of developing adverse reactions under Dose A to the odds of developing adverse reactions under Dose B:

$$\frac{\left(\frac{20}{30}\right)}{\left(\frac{40}{50}\right)} = 0.8\overline{3} = \frac{1}{1.2}$$

The odds of developing adverse reactions under Dose B is 1.2 times the odds of developing an adverse reaction under Dose A.

(e) **Without doing any additional calculations**: Consider the quantity "the odds ratio comparing the odds of not developing adverse reactions under Dose B to the odds of not developing adverse reactions under Dose A." How would this quantity compare to the value you calculated in Part (d) above

- √ **It would be the same**
- ◯ The quantity in this part would be the reciprocal of the quantity in Part (d) above

Solution: Compared to Part (d), we've swapped the notion of success (from "developing adverse reactions" to "not developing adverse reactions") and the baseline (from Dose A first to Dose B). Therefore, we should arrive back at the original odds ratio. If you don't believe me:

- Odds of not developing adverse reactions under Dose B: 50/40
- Odds of not developing adverse reactions under Dose A: 30/20
- Odds ratio comparing the odds of not developing adverse reactions under Dose B to the odds of not developing adverse reactions under Dose A:

$$\frac{\left(\frac{50}{40}\right)}{\left(\frac{30}{20}\right)} = 0.8\overline{3}$$

(f) **Without doing any additional calculations**: Consider the quantity "the odds ratio comparing the odds of not developing adverse reactions under Dose A to the odds of not developing adverse reactions under Dose B." How would this quantity compare to the value you calculated in Part (d) above?

◯ It would be the same

√ **The quantity in this part would be the reciprocal of the quantity in Part (d) above.**

Solution: Compared to Part (d), we've swapped the notion of success (from "developing adverse reactions" to "not developing adverse reactions") but left the notion of the baseline unchanged. Therefore, we should arrive at the reciprocal of the original odds ratio.

If you don't believe me:

- Odds of not developing adverse reactions under Dose A: 30/20
- Odds of not developing adverse reactions under Dose B: 50/40
- Odds ratio comparing the odds of developing adverse reactions under Dose A to the odds of developing adverse reactions under Dose B:

$$\frac{\left(\frac{30}{20}\right)}{\left(\frac{50}{40}\right)} = 1.2 = \frac{1}{0.8\overline{3}}$$